

ELLENKI VARSHITHA

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Hyderabad, India

SUMMARY

Backend-focused software developer with hands-on experience building production-grade systems using Java, Spring Boot, and microservices architecture. Proficient in applied AI and RAG pipelines, having built an agentic codebase explainer leveraging Claude, HuggingFace embeddings, and pgvector. Solved 300+ DSA problems and shipped full-stack, containerised projects across job portals, e-commerce backends, and NLP systems. Eager to contribute to high-impact backend engineering teams.

TECHNICAL SKILLS

Languages: Java, Python, C, JavaScript, SQL, HTML5, CSS3

Frameworks & Libraries: Spring Boot, Spring MVC, Spring Security, React.js, Node.js, Express.js, Flask, FastAPI, Tailwind CSS

APIs & Architecture: REST APIs, GraphQL, WebSockets, Microservices, MVC, JWT, OAuth 2.0

Databases: MySQL, PostgreSQL, MongoDB, Redis, SQLite, Amazon DynamoDB, Neo4j, BigQuery

AI / ML: Scikit-learn, Pandas, NumPy, HuggingFace, LLM APIs, RAG, Model Context Protocol (MCP)

Tools: Git, GitHub, Docker, Postman, Swagger/OpenAPI, Maven, Jira, Vercel, Netlify

Cloud: AWS (EC2, S3, RDS, IAM), Google Cloud Platform, Microsoft Azure

CS Fundamentals: Data Structures & Algorithms (300+ problems), OOP, DBMS, System Design, OS

EDUCATION

St. Martins Engineering College

B.Tech, Artificial Intelligence and Data Science

Hyderabad, India

Aug 2022 – May 2026

- Relevant Coursework: Data Structures & Algorithms, Databases, Computer Networks, Operating Systems, Machine Learning

PROJECTS

CodeLens | *Spring Boot, RAG, Claude API, HuggingFace, pgvector, FastAPI, Docker* 2025 – 2026

- Built an agentic AI codebase explainer that clones GitHub repositories and makes large codebases queryable via natural language — solving the core challenge of repos being too large for LLM context windows.
- Implemented a RAG pipeline: chunked code files, generated embeddings with HuggingFace, stored vectors in pgvector as a distributed vector storage and index system, and retrieved semantically relevant chunks to pass as context to Claude with low-latency query performance.
- Exposed a FastAPI REST interface; containerised with Docker for portability and ease of deployment.

CareerNest | *Java, Spring Boot, Spring Security, JWT, PostgreSQL, React.js, Docker, Vercel* 2025

- Developed a full-stack job portal with role-based access control for Recruiters, Applicants, and Admins using Spring Boot and JWT authentication, exposing 8+ secure REST endpoints consumed by a React frontend.
- Designed a normalised PostgreSQL schema with 6 entities using Hibernate/JPA; implemented advanced job search with filters, pagination, and resume upload; containerised with Docker and deployed on Vercel.

ShopSphere | *Java, Spring Boot, Spring Cloud, GraphQL, MySQL, Redis, RabbitMQ, Docker* 2024 – 2025

- Architected a fault-tolerant microservices e-commerce backend with 5 independent Spring Boot services (User, Product, Cart, Order, Notification) communicating via REST and async RabbitMQ events through a Spring Cloud API Gateway; designed for high-volume traffic and horizontal scalability.
- Exposed a GraphQL endpoint on the Product service and added Redis caching for low-latency reads, reducing average response time by ~60% under 100 concurrent users; documented all APIs with Swagger/OpenAPI.

OPEN SOURCE

GirlScript Summer of Code (GSSoC)

Open Source Contributor

Remote

May 2026 – Present

- Contributed to 4 repositories including wagtail/wagtail — raised 5 PRs with 3 merged; worked across Python, TypeScript, and JavaScript codebases fixing UI bugs, improving documentation, and adding features.

CERTIFICATIONS

NPTEL Cloud Computing — Elite + Silver

Score: 81%

IIT Kharagpur

Jan 2025 – Apr 2025